

LEGAL AND ECONOMIC DIMENSIONS OF REAL-WORLD ASSET (RWA) TOKENIZATION: GOLD AS TOKENIZED ASSET

DIMENSÕES JURÍDICAS E ECONÔMICAS DA TOKENIZAÇÃO DE ATIVOS DO MUNDO REAL (RWA): O OURO COMO ATIVO TOKENIZADO

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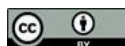
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Abstract

With rapid technological advancements, the business and financial sectors are undergoing significant transformation, and blockchain has emerged as a key driver in disrupting traditional financial ecosystems while promoting greater financial inclusion, particularly in countries like Indonesia that are in the midst of transitioning toward a digital economy. One of blockchain's core capabilities is asset tokenization, which has attracted growing interest from financial institutions seeking new investment mechanisms. This study uses a qualitative approach and descriptive analysis through a literature review to examine the potential and challenges of tokenization, using the tokenization of real-world assets (RWA), specifically gold, as an illustrative example. The findings reveal that current development of tokenization is promising, as it enhances liquidity, operational efficiency, and global accessibility, while blockchain, as its core technology, provides greater transparency and enables fractional ownership. However, several issues remain as the tokenization of real-world assets gains momentum, including legal uncertainty, as governments and institutions adapt to the pace of new financial innovations while seeking to protect consumers and promote innovation. The study recommends adopting market-driven regulatory frameworks and encouraging joint efforts with international

Resumo

Com os rápidos avanços tecnológicos, os setores empresarial e financeiro estão passando por transformações significativas, e a blockchain emergiu como um fator-chave na ruptura dos ecossistemas financeiros tradicionais, promovendo maior inclusão financeira, particularmente em países como a Indonésia, que estão em plena transição para uma economia digital. Uma das principais capacidades da blockchain é a tokenização de ativos, que tem atraído crescente interesse de instituições financeiras em busca de novos mecanismos de investimento. Este estudo utiliza uma abordagem qualitativa e análise descritiva por meio de uma revisão da literatura para examinar o potencial e os desafios da tokenização, usando a tokenização de ativos do mundo real (RWA), especificamente o ouro, como exemplo ilustrativo. Os resultados revelam que o desenvolvimento atual da tokenização é promissor, pois aumenta a liquidez, a eficiência operacional e a acessibilidade global, enquanto a blockchain, como sua tecnologia central, proporciona maior transparência e permite a propriedade fracionada. No entanto, vários problemas permanecem à medida que a tokenização de ativos do mundo real ganha impulso, incluindo a incerteza jurídica, visto que governos e instituições precisam se adaptar ao ritmo das novas inovações financeiras, buscando proteger os consumidores e promover a inovação. O



institutions to compare and evaluate market practices across different jurisdictions.

Keywords: Blockchain, Tokenization, Real-World Asset, Technology, Finance.

estudo recomenda a adoção de estruturas regulatórias orientadas pelo mercado e o incentivo a esforços conjuntos com instituições internacionais para comparar e avaliar as práticas de mercado em diferentes jurisdições.

Palavras-chave: Blockchain. Tokenização. Ativo do Mundo Real. Tecnologia. Finanças.

1 INTRODUCTION

The rise of blockchain technology in Indonesia is an undeniable phenomenon. In an era of globalization and digitalization, Indonesia is rapidly aligning with global technological trends, driven largely by its young, digitally literate population. Blockchain-based digital assets, initially developed abroad, have increasingly become embedded within Indonesian society, supported by expanding internet access and the growing affordability of electronic devices such as computers and smartphones. This development is particularly significant due to the emergence of decentralized financial systems, which allow open participation and collective validation of transactions. Through distributed verification mechanisms, blockchain ensures transparency by recording each process immutably within the electronic system.

Fintech developments today reflect the principles of Architecture 4.0, a design approach that is fundamentally data-driven and fully interconnected through the internet. Built upon real-time data, cloud computing, and blockchain infrastructure, fintech delivers enhanced efficiency, practicality, and accessibility in financial services¹. Traditional financial processes are increasingly transformed into integrated digital ecosystems, enabling seamless interactions, reduced operational frictions, and broader financial inclusion.

Asset tokenization represents a transformative development in accounting and asset management, whereby assets are converted into programmable digital tokens. This process creates additional liquidity by enabling the circulation of assets that were previously illiquid or underutilized. Tokenization enhances transparency by clearly recording ownership rights and transaction histories, while also increasing efficiency by

¹ Napitupulu, Diana.R.W, *Kamus Financial Technology* (Jakarta: UKI Press, 2023), 9.

reducing costs associated with asset management, issuance, and intermediaries². The tokenization of real-world assets such as gold, property, and bonds has become a central topic in global financial discourse. At the World Economic Forum Annual Meeting 2024, this theme was highlighted as a major opportunity for business and financial markets, emphasizing how blockchain-based digital representations of physical and financial assets can be securely exchanged in real time³. Fundamentally, blockchain strengthens the value of time-stamped records by addressing inefficiencies arising from poor information quality through cryptographic hashing and immutable transaction records.⁴

According to Pablo Hernández de Cos, economist and former Governor of the Bank of Spain, crypto-assets can be classified based on the primary nature of the token itself. Tokens are discrete units of digital value issued to facilitate interactions within a virtual environment. Some, known as native tokens, are created independently of any underlying asset and are intrinsically linked to a specific distributed ledger. Bitcoin, which is inseparable from its blockchain network, exemplifies this category. The value of such tokens is largely driven by users' expectations of future utility rather than underlying cash flows or issuer fundamentals⁵. As token utility continues to evolve, blockchain communities worldwide are increasingly exploring tokenization for both tangible and intangible assets.

Following years of investment, proof-of-concept initiatives, and extensive testing, the tokenization of financial assets is now occurring at institutional and governmental levels. This transition is expected to reshape national trading systems and promote more inclusive financial participation. However, rapid innovation has also introduced complex economic and legal challenges. Governments globally have recognized the urgency of regulating digital assets due to their accelerated growth. In Indonesia, digital financial assets and related innovations classified as Innovation Technology in the Financial Sector (ITSK) are regulated under the Law on Financial Sector Development and Strengthening

² Andryushin S.A., Tokenization of real assets: classification, platforms, applications, opportunities and challenges of development. *Russian Journal of Economics and Law*. 2024;18(1):88-104. (In Russ.)

³ Rooz, Yuval, Financial and Monetary Systems: How tokenization is transforming global finance and investment. World Economic Forum, 2024. <https://www.weforum.org/stories/2024/12/tokenization-blockchain-assets-finance/>

⁴ Casey, Michael J and Vigna, Paul, *The Truth Machine, the Blockchain and the Future of Everything*. (New York: St. Martin's Press, 2018). 164 – 166.

⁵ Pablo Hernández de Cos, "Crypto-assets: a financial authority's view". Bank of International Settlements, Masters in Finance Closing of the Speakers Series, ESADE (31 May 2022). <https://www.bis.org/review/r220621c.pdf>

(UU P2SK). This law designates the Indonesian Financial Services Authority (OJK) as the primary regulatory body, granting it a central role in overseeing and advancing tokenization initiatives.

For the purposes of this research, real-world asset tokenization is examined through three investment classes: gold, property, and government bonds. With tokenization models for these assets having received OJK approval, particular emphasis is placed on gold tokenization. Gold's enduring role as a tangible safe-haven asset, driven by its scarcity and diverse applications, makes it a compelling case study. This focus coincides with a significant regulatory milestone, as OJK continues to develop an adaptive and inclusive governance framework to support the sustainable growth of asset tokenization in Indonesia.

This study is subject to several limitations. First, the analysis is constrained by the evolving nature of Indonesia's regulatory framework, which may continue to change as tokenization technologies mature. Second, the research focuses primarily on gold, property, and government bond tokenization, thereby excluding other emerging asset classes that may also hold significant relevance. Finally, empirical data on large-scale implementation remains limited due to the relatively early stage of institutional adoption, which may affect the generalizability of the findings.

2 METHODS

The study employed a normative legal research method to analyze the current state of blockchain adoption in Indonesia's business and financial sectors. Real-World Asset tokenization using blockchain technology is now regulated by OJK and is beginning to be adopted by various organizations as practical use cases, notably gold, one of the most favored investment assets among Indonesians has become a key focus, with gold tokenization identified as one of the three priority areas of tokenization regulated by the OJK. A normative legal research method is a scientific procedure to discover truth based on the logical framework of legal science from a normative perspective, relying on consistent scientific reasoning rooted in the discipline and working methods of normative legal science, whose primary object of study is the law itself.⁶ In normative legal

⁶ Tehupeiori, Aarce, *Bahan Ajar Metode Penelitian Hukum*. (Jakarta: UKI Press, 2021): 55.

research, legal sources consist of three sources which are primary legal materials consisting of legislation and official records, secondary legal materials include all published legal materials and tertiary legal materials to provide explanations to support secondary legal materials, such as dictionaries. Using Richard Posner's theory of Economic Analysis of Law as an analytical tool, the analysis examines, compares and contrasts the development of tokenization as an investment asset notably gold, in Indonesia and the development of tokenization of gold globally. The analysis discusses global challenges that are relevant in Indonesia as technology advances and shape the way businesses and the financial sector operate.

3 RESULTS AND DISCUSSION

3.1 Blockchain tokenization in the business and financial sectors

When people hear the term blockchain, they often associate it solely with crypto assets or Bitcoin. In reality, blockchain is a broader technology that functions as a digital ledger capable of securely storing data and retrieving it through cryptographic hash values. Blockchain operates as an immutable, distributed ledger that records transaction details in blocks; each new transaction is added as a new block to the existing chain and is validated through a consensus mechanism among network participants. As a result, altering data on the blockchain would require immense computational power and simultaneous modification of all distributed copies of the ledger, making unauthorized manipulation extremely difficult. In addition, blockchain enables smart contracts which is essentially a self-executing logic embedded within the network that automates transactions and processes. While blockchain is decentralized rather than centrally controlled, it is not entirely self-sustaining and relies on participants who choose to operate and maintain the network.⁷ Therefore, it is safe to say that blockchain is a type of distributed ledger technology (DLT) that stores data across a decentralized network rather than relying on traditional centralized systems, which are more vulnerable to data loss, security breaches, and operational inefficiencies. It offers several key advantages. Greater transparency is achieved because the entire lifecycle of the ledger is visible to

⁷ Shrivastava, Gulshan; Le, Dac-Nhuong ; Sharma, Kavita, *Cryptocurrencies and Blockchain Technology Applications*. (New Jersey: John Wiley & Sons, Inc, 2020), 4.

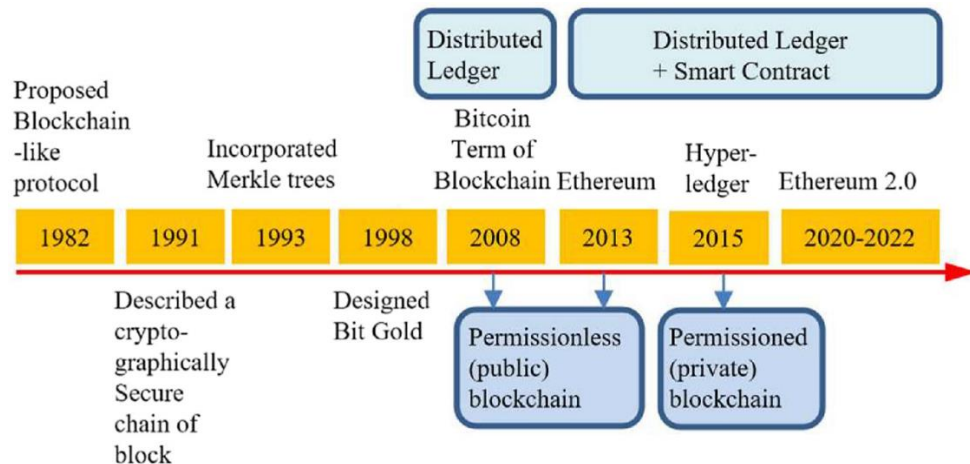
authorized participants. Cost savings arise from shared infrastructure, reducing the need for duplicative processes such as repeated data entry and reconciliation, enabling all parties to save time and resources. Operational efficiency is enhanced through smart contracts, which automate and synchronize transactions, allowing organizations to redirect their efforts toward higher-value activities. Finally, enhanced security is ensured through cryptographic techniques that encrypt transactions and maintain an immutable record of all activity within the digital ledger. Blockchain has emerged as a response to longstanding structural weaknesses in the business and financial sectors, as it offers a transformative technological shift: its decentralized, transparent, and tamper-resistant architecture can strengthen trust, enhance auditability, and reduce information asymmetry across financial markets. By enabling real-time verification, immutable transaction records, and automated compliance through smart contracts, blockchain directly addresses many of the systemic vulnerabilities that contributed to past crises, therefore positioning it as a critical innovation for building a more resilient and accountable financial ecosystem.

In 1982, David Chaum became the first known figure to propose a blockchain-like protocol in his Ph.D. thesis, laying the groundwork for later advancements. Subsequent research followed, including Nick Szabo's 1998 design of "bit gold," a decentralized digital currency mechanism. A major breakthrough came in 2008 when Satoshi Nakamoto introduced Bitcoin, a peer-to-peer electronic cash system, and the term blockchain was first used to describe the distributed ledger supporting Bitcoin transactions. In 2013, Vitalik Buterin proposed Ethereum in his whitepaper. The project was crowdfunded in 2014, and the Ethereum network officially launched on July 30, 2015, marking the beginning of Blockchain 2.0. Unlike earlier blockchain projects that primarily focused on creating alternative cryptocurrencies, Ethereum enabled trustless distributed applications and introduced smart contracts—self-executing computer programs stored on the blockchain. Its upgrade to Ethereum 2.0, rolled out between 2020 and 2022, aimed to significantly improve speed, scalability, efficiency, and security. In 2015, the Linux Foundation launched the Hyperledger project, an open-source initiative designed to build enterprise-grade blockchain frameworks. Unlike public networks such

as Bitcoin and Ethereum, Hyperledger supports permissioned blockchains tailored for business use cases.⁸ The history of blockchain is summarized in Fig. 1.

Figure 1

History of Blockchain



Source: A survey on blockchain technology and its security⁹

Figure 1 illustrates the chronological evolution of blockchain technology, beginning in the early 1980s with the proposal of blockchain-like protocols and the cryptographic foundations of secure, chained blocks in the early 1990s, including the introduction of Merkle trees. The concept matured with the emergence of Bitcoin in 2008, which introduced a distributed, permissionless public ledger and marked the first practical implementation of blockchain. This was followed by Ethereum in 2013, which expanded blockchain functionality through smart contracts, enabling programmable and decentralized applications. Subsequent developments led to hyper-ledgers and permissioned (private) blockchains around 2015, designed to meet enterprise and institutional needs. The evolution culminates with Ethereum 2.0 (2020–2022), which integrates advanced distributed ledger architectures and smart contracts with improvements in scalability, efficiency, and governance, reflecting the growing sophistication and diversification of blockchain systems.

Blockchain has emerged as one of the most prominent disruptive innovations, attracting significant scholarly attention. Stakeholders, including developers,

⁸ Huaqun Guo, Xingjie Yu “A survey on blockchain technology and its security”, *Blockchain: Research and Applications* 3 (2022) 100067. DOI 10.1016/j.bcra.2022.100067

⁹ Guo & Yu (2022)

entrepreneurs, and technology advocates argue that blockchain has the capacity to reshape economic, legal, political, and cultural systems. By contrast, critics contend that both the concept and its practical applications remain unclear and uncertain. Academic inquiry into blockchain, Bitcoin, and related initiatives began to gain momentum within the business literature around 2014.¹⁰

Businesses today differ greatly from those of the 20th century, when technology merely served as an optional tool and organizations could still choose not to adopt it. Similarly in the financial sector, investment assets were traditionally shaped by market needs and sentiment, and as technology progressed in the 1980 - 1989 period, the closed system approach to finance began expanding toward broader environments within an open-system network. The long-standing paradigm focused solely on shareholders' wealth maximization underwent a Kuhnian shift, moving beyond the boundaries of normal science and evolving into a multivalued framework that incorporated social criteria and other considerations. This transformation is reflected in the application of the financial principles to a wider array of organizations beyond traditional business firms, signaling a more holistic understanding of value and organizational purpose.¹¹ In today's digital era, businesses no longer have the option to ignore technological advancements and the financial sector is increasingly shaped by the demand for higher transaction volumes, transparency, speed, efficiency and minimal margin of errors. Since the turn of the 21st century, technological advancement has accelerated even more rapidly than in the previous century, giving rise to innovations that are reshaping industries across sectors and around the globe. In 2008, the global financial crisis driven by mortgage backed securities and complex derivative products revealed how opacity, inadequate risk assessment, and limited transparency can destabilize entire economies. The 2008 crisis exposed how opaque derivatives, such as mortgage-backed securities and credit default swaps, amplified systemic risk due to poor asset traceability, inadequate collateral management, off-balance-sheet exposures and fragmented recordkeeping. The systemic failures that surfaced during this period prompted the adoption of stricter regulatory frameworks, including Basel II, which strengthened standards for risk management,

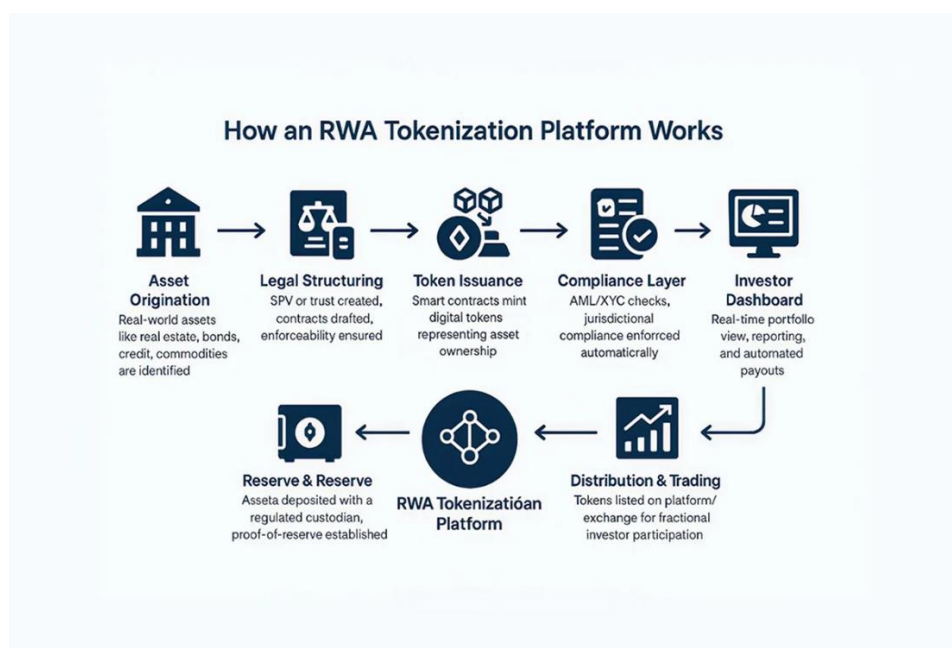
¹⁰ Julie Frizzo-Barker, Peter A. Chow-White, Philippa R. Adams, Jennifer Mentanko, Dung Ha, Sandy Green, "Blockchain as a disruptive technology for business: A systematic review". *International Journal of Information Management*. Volume 51 (April 2020). <https://doi.org/10.1016/j.ijinfomgt.2019.10.014>

¹¹ Rahnema, Ahmad, "Latest Developments in the Field of Finance Latest Developments in the Field of Finance Introduction 1" Working Paper WP-188 IESE Business School – University of Navarra (May, 1990).

capital adequacy, and market discipline. Basel II was introduced to address weaknesses in earlier capital frameworks by making banks more risk-sensitive as it emphasized in improving risk measurement, capital adequacy linked to actual exposure and Market discipline through disclosure. Furthermore, for complex financial instruments, particularly structured products and derivatives that require better visibility, valuation, and governance, Basel II laid the conceptual groundwork for financial systems that demand traceability, accurate risk attribution, and real-time reporting¹². What is then required is a new mindset and a shift toward infrastructure-driven trust, where risk is managed not only by regulation, but by how financial instruments are issued, recorded, and settled. The 21st century pushed forward the use of blockchain technology and the concept of tokenization continues to develop, taking shape as Real-World Asset (RWA) tokenization.

Figure 2

How an RWA Tokenization Platform Works



Source: Antiers Solutions¹³

¹² Bank of International Settlement. Consultative Paper: Strengthening the resilience of the banking sector (17 December 2009) <https://www.bis.org/publ/bcbs164.htm>

¹³ RWA Tokenization: Practical Guide for Investors, Builders, and Institutions <https://www.antiersolutions.com/blogs/rwa-tokenization-practical-guide-for-investors-builders-and-institutions/>

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RWA tokenization is the process of converting legally enforceable rights to a real-world asset into blockchain-based tokens that can be transferred, fractionalized, and integrated into digital financial systems. The token does not constitute the asset itself; it merely represents the rights associated with that asset. Without a supporting legal structure, these tokens are simply metadata recorded on a blockchain, lacking legal force or enforceability.

The above figure shows the process of RWA and that the value of RWA tokenization is fundamentally derived from efficiency gains. Its real strength lies in providing structural levers that reduce costs, expand investor access, and enhance liquidity. When designed effectively, these efficiencies translate into measurable ROI improvements for both issuers and investors. However, every efficiency benefit also introduces new risks. For platform builders, identifying these vulnerabilities early and incorporating safeguards into system design is essential for building a sustainable tokenization model rather than a short-lived pilot. RWA tokenization delivers clear advantages to both investors and issuers, reinforcing its credibility as a modern financial infrastructure. While it shares similarities with traditional securitization, both fractionalize and distribute claims on underlying asset, that the key distinction lies in the underlying technology. Tokenization leverages blockchain infrastructure to offer greater transparency, accessibility, and operational efficiency than conventional securitization frameworks. In retrospect, derivatives evolved to transfer and manage risk, but their

complexity and opacity became a source of instability. Tokenized real-world assets represent a partial reversal of the earlier trajectory toward synthetic financial exposure. Whereas complex derivatives often detach investments from their underlying assets, RWA tokenization reanchors value in direct, verifiable asset backing. By embedding legal rights and compliance mechanisms into digital representations, tokenization enables regulators and institutions to monitor exposures continuously rather than retrospectively. In this way, it offers a new financial architecture that operationalizes the core principles of Basel II and post-crisis reforms, aiming to make financial markets safer by design rather than relying solely on supervisory oversight.

Property does not consist solely of physical assets such as land and buildings; it also encompasses the legal aspects, rights, and economic benefits arising from ownership of those assets. The property business is therefore highly attractive to investors, as it represents a relatively sustainable form of investment with strong long-term potential. The consistent appreciation of property values over time further enhances its appeal by offering steady capital gains for investors.¹⁴ These characteristics make real estate particularly suitable for tokenization, as the underlying rights and interests in property can be digitally represented, divided, and transferred through blockchain-based tokens. Real estate tokenization is the process of converting the value of a physical property into digital tokens that can be bought, sold, or traded on a blockchain platform, whereby each token represents fractional ownership, thus allowing investors to participate in real estate markets without needing substantial capital and the benefit of efficient transactions as tokenization streamline the transaction process, reducing costs and accelerating the buying and selling of property. This innovative model not only broadens access to real estate investments but also enhances market liquidity and transparency. Real Estate tokenization has several advantages, that it enables fractional ownership, making it easier for investors to enter or exit positions as opposed to traditional real estate that requires large capital investments upfront and takes time to sell if needed. Transparency and Security offered by Blockchain technology ensures secure, verifiable, and transparent records of ownership and transactions, therefore improves transparency across the real estate sector and ultimately has the potential to unlock significant global investment

¹⁴ Harjono, Dhaniswara.K, *Hukum Properti* (Depok: Rajawali Press, 2019), v – 2.

opportunities¹⁵. According to the research done by The Deloitte Center for Financial Services, their prediction that US\$4 trillion of real estate will be tokenized by 2035, increasing from less than US\$0.3 trillion in 2024, with a CAGR of 27%, based on three underlying components of the global forecasted tokenized real estate ecosystem: tokenized private real estate funds, tokenized ownership of loans and securitizations, and tokenized ownership of undeveloped land or under-construction projects.¹⁶

Because tokenized assets are fractionalized, businesses must adopt new valuation methods that accurately reflect this structure. Real estate tokens can also be leveraged by financial institutions and investment platforms to develop innovative financial products, including tokenized property funds or hybrid portfolios that combine digital and traditional asset classes. This diversification strategy addresses liquidity challenges in real estate finance while expanding investment opportunities. In addition, these insights can guide policymakers and regulators in formulating robust legislative frameworks that support the growth of tokenized real estate markets while ensuring investor protection and compliance with know-your-customer (KYC) and anti-money laundering (AML) requirements. To foster confidence and stability in the industry, establishing standardized procedures for smart contract governance, asset verification, and secondary market regulation is essential.¹⁷

The United Arab Emirates, especially Dubai and Abu Dhabi, has emerged as a leader in the adoption of blockchain technology and tokenized assets, through initiatives such as the Virtual Assets Regulatory Authority (VARA) and the Abu Dhabi Global Market's distributed ledger technology frameworks, the UAE offers a clear and transparent regulatory environment for the issuance of digital assets, notably for real-estate token. VARA's licensing regime provides legal certainty and regulatory clarity for tokenization initiatives as robust investor appetite for digital real estate and blockchain-enabled property transactions continues to support the growth of the tokenized asset ecosystem. Germany is one of the first EU countries to legally recognize blockchain-

¹⁵ Nagayach, Mayank, *Real estate tokenization: A new era for property investment and Luxembourg's strategic role*. Luxembourg: EY, 2025.

¹⁶ John D'Angelo, Jeffrey J. Smith, Tim Coy, Parul Bhargava, *Digital dividends: How tokenized real estate could revolutionize asset management*. (United States of America: Deloitte Center for Financial Services, 24 April 2025).

¹⁷ el Jaouhari A, Samadhiya A, Chokshi H, Kumar A, "Tokenization and the future of property investment: A new paradigm for real estate" *International Journal of Strategic Property Management* 29(4) (August 2025). DOI:10.3846/ijspm.2025.24814

based securities. The *Electronic Securities Act (eWpG)* and BaFin regulations allow full compliance for tokenized assets. Switzerland, through the Swiss Financial Market Supervisory Authority (FINMA), provides a solid legal framework and clear guidance for token issuance, even provide tax benefits for digital asset businesses

Another example of the use case of Real-World Asset tokenization used by businesses is bonds. Tokenized bonds are rapidly emerging as one of the most prominent asset classes within the expanding digital asset ecosystem. Tokenized bonds have evolved from traditional debt instruments to a fixed-income featuring blockchain technology, tokenized bonds offer faster transactions, enhanced liquidity, and lower issuance costs. Both business entities and financial institutions need to understand the key legal, operational, and technological considerations required for a successful tokenized bond issuance, including ownership recording, custodial arrangements, and servicing processes. According to Hong Kong Monetary Authority through their Research Memorandum in November 2023, bond tokenisation is still in its early stage of development. Nevertheless, some Asian and European institutions have already taken an early step to embrace this wave of technological advancement, by applying tokenisation to their bond issuance and developing digital platforms for this purpose. At the end of March 2023, our data reveal that the total issuance amount of tokenised bonds globally had reached US\$3.9 billion, with nine-tenths of the issuance occurring in and after 2021.¹⁸

On 25 July 2024, the Republic of Slovenia issued its first digital bond, making it the first EU sovereign and among the first globally to complete such a transaction. The issuance took place within the European Central Bank's wholesale central bank money settlement pilot, with the bond settled on-chain using wholesale CBDC through the Banque de France's DL3S interoperable tokenized cash system. BNP Paribas served as Global Coordinator, Sole Bookrunner, and operator of the Neobonds DLT platform, a private tokenization system built with Digital Asset's Daml and running on the Canton blockchain. The digital bond carried a nominal value of EUR 30 million, a 3.65% coupon, and a maturity date of 25 November 2024. Slovenia has expressed a strong commitment to adopting new technologies in its central government debt management and across its broader financial market infrastructure. These early tokenization initiatives and wholesale CBDC settlement trials are viewed as important steps toward enhancing transparency and

¹⁸ Hong Kong Monetary Authority, An Assessment on the Benefits of Bond Tokenisation, Research Memorandum 04/2023, (Hong Kong: 28 November 2023)

efficiency in financial markets as technological adoption expands. Although the current market size for such instruments remains small, the role of distributed ledger technology is expected to grow substantially in the coming years.¹⁹

The evolution of real-world asset (RWA) tokenization has expanded beyond tangible assets to include intangible assets such as copyrights and intellectual property. This development extends the economic rights of creators and copyright holders into the blockchain ecosystem, enabling those rights to be recognized, represented, and exercised through tokenization mechanisms. Economic rights constitute the exclusive entitlement of creators or copyright holders to derive financial benefits from their copyrighted works. Law of the Republic of Indonesia Number 28 of 2014 on Copyright, Article 8 of the Copyright Law establishes both the rationale and objective of these rights, ensuring that creators are able to capture the economic value generated by their creations. By contributing original works, creators enhance cultural development and deliver meaningful value and enrichment to society. Accordingly, it is both fair and appropriate that those who benefit from such creative works provide compensation as a form of recognition to the creators or copyright holders.²⁰

An analysis of the legal and economic dimensions of RWA tokenization can be effectively framed through the Economic Analysis of Law, particularly Richard Posner's theory, which evaluates legal rules based on their capacity to promote economic efficiency and social welfare. From this perspective, the transparency inherent in tokenization, where ownership records and transaction histories are immutably recorded on a blockchain, directly aligns with Posner's objective of reducing information asymmetry, a key source of market inefficiency. By enabling market participants and regulators to access reliable, real-time information, tokenization lowers transaction costs, mitigates opportunistic behavior, and enhances allocative efficiency.

Accordingly, a legal framework that accommodates and supports tokenization does more than enable technological innovation; it advances the core economic function of law as envisioned by Posner. Clear, adaptive, and efficiency-oriented regulations can maximize the benefits of tokenization by fostering openness, expanding market participation, and reducing friction in economic activity. Under Posner's approach, such

¹⁹ The Republic of Slovenia, Ministry of Finance. The Republic of Slovenia issues its inaugural digital bond (July 2024). <https://www.gov.si/en/news/2024-07-26-the-republic-of-slovenia-issues-its-inaugural-digital-bond/>

²⁰ Nainggolan, Bernard, *Komentar Undang-Undang Hak Cipta* (Bandung:2016). Hlm. 62

a regulatory framework has the potential to improve overall social welfare by ensuring that legal institutions evolve in tandem with technological developments that enhance efficiency and transparency in financial markets.

3.2 Real-world asset tokenization: gold

Gold is a highly valuable metal with a wide range of applications across various sectors. It is widely used in jewelry and metalwork, as well as in computer processors and other electronic components such as semiconductors, printed circuit boards, and electrical contacts. In the medical and healthcare fields, gold is utilized in dental fillings and radiographic procedures. Beyond its industrial and technological uses, gold also serves as an important financial instrument, playing a critical role in the global economy—particularly as a form of currency reserve in the form of bullion or coins. For centuries, gold has served as a store of value, a medium of exchange as well as social symbol in most of societies in the world. During wars, gold has always been used to preserve individual wealth and because of its scarcity, durability and broad social acceptance as “safe haven” money, investment in gold gives cushion to inflation, currency crises and throughout times of uncertainty to preserve purchasing power. Gold is especially important to understand because of its close connection in popular and economic theories with money. For economists and political economists alike, gold is a natural money substance due, again, to its physical qualities and its natural scarcity. Its conversion into money enables the expansion of trade and the generalization of market economies²¹.

Gold has been invested in many ways and according to one’s preference and need. Physical bullion such as bars, coins are direct ownership but requires logistics, secure storage and insurance. To a certain extent, gold in a form jewelry represents investment value, cultural value and even aesthetic value, however it would also carries logistics, secure storage and insurance in additional to design and manufacturing costs. Financial institutions offer financial products backed by physical gold namely Exchange Traded Fund (ETF), Futures and options with gold underlying, gold mining equities and funds which gives exposure to producers with operational risk and leverage to metal prices. Sovereign gold bonds and government schemes would offer stable interest.

²¹ Schoenberger, Erica, “Why is Gold Valuable? Nature, Social Power and the Value of Things” *Cultural Geographies* 18(1) (January 2011):3-24 DOI:10.1177/1474474010377549

Gold has become one of the most lucrative businesses for banks, taking example of PT Bank Syariah Indonesia (BSI), the country's largest Islamic bank, reported a sharp 78.17% year-on-year increase in its gold business, reaching Rp 12.8 trillion (USD 783 million) in 2024, this growth is due to consistently rising global gold prices, which have increased by 20–30% annually and strengthened gold's position as a safe-haven asset amid economic uncertainty. The bank is aggressive in promoting its gold-related businesses namely gold pawn and installment services, emphasizing gold's role as an inflation hedge and supporting public education on safe gold investment amid the rise of fraudulent schemes and interestingly, the bank notes that nearly half of its gold installment customers are millennials and Gen Z, reflecting growing interest among younger investors²².

Beyond investment and jewelry, gold has important industrial uses because of its excellent electrical conductivity and corrosion resistance: connectors, small wires, plated contacts, microelectronics, some medical devices and precision instruments. Small quantities of gold are present in nearly all smartphones, computers and many telecom/electrical connectors, which creates recurring industrial demand. Its extensive usefulness stems from the unique combination of its physical and chemical properties. Gold is among the densest metals, naturally lustrous, exceptionally durable, soft, and highly resistant to corrosion. It also offers excellent electrical and thermal conductivity and is the most malleable and ductile element known, making it invaluable across industries and financial systems alike.

Given the existing technology of blockchain, tokenization of gold gives opportunities and benefits as it allows fractional ownership therefore it provides financial inclusion by widening investor participation. Liquidity is improved as tokenized units can trade 24/7 on digital markets, making traditionally illiquid gold bullion become easier to buy and sell. Tokenization allows on-chain transfers and programmable rules (smart contracts) can reduce reconciliation, custody and settlement costs, hence increases efficiency and lowers overall cost and processing time. Transparency and provenance are provided as tokens can carry on-chain metadata of the origin and vault location of the tokenized gold, improving auditability when properly implemented.

²² Kasanjanu, Bagus and Nuky, Ester, "BSI's Gold Business Surges 78%, Reaching IDR 12.8 Trillion" Investor Trust, 3 February 2025 <https://investortrust.id/financial/55231/bsis-gold-business-surges-78-reaching-idr-128-trillion?page=english-edition>

Current market leaders of gold tokens are Tether and Paxos. Tether was established by a group of early Bitcoin adopters and digital currency advocates with the aim of enabling the use of fiat currencies in a digital form. In 2014, the company launched Tether.to, a blockchain-based platform designed to modernize traditional financial systems by tokenizing fiat currencies. These include USD-pegged Tether Tokens (USD₮), as well as tokens pegged to the euro (EUR₮), Chinese yuan offshore (CNH₮), and Mexican peso (MXN₮). Beyond fiat currencies, Tether has also expanded into the tokenization of commodities, notably gold. Tether Gold (XAU₮), launched in 2020 by TG Commodities, S.A. de C.V., is a gold-backed stablecoin that combines the advantages of physical gold with the flexibility of digital assets. Each XAU₮ token represents ownership of one fine troy ounce of physical gold held in custody and compliant with the London Bullion Market Association (LBMA) Good Delivery standard. The product enables investors to gain exposure to physical gold without the traditional disadvantages of physical ownership, such as high storage costs and limited accessibility. Tether Gold does not charge ongoing custody fees, applying only a one-time fee of 25 basis points at the time of purchase or redemption. Notably, XAU₮ tokens are divisible up to six decimal places, allowing fractional ownership down to 0.000001 troy ounce of gold.²³ According to Bloomberg report, Tether Holdings SA, while the company is registered in El Salvador, it has its own vault in Switzerland to hold an \$8 billion stockpile of gold.²⁴

PAXG is a crypto asset token directly backed by physical gold, with each token representing one troy ounce of London Good Delivery gold, securely stored in vaults in The token is issued by Paxos Trust Company and operates on the Ethereum blockchain under the ERC-20 standard. Unlike fiat-backed stablecoins such as USDT, the value of PAXG is pegged to the global price of gold. Secure and regulated. The allocated gold that backs PAXG is custodied in London Bullion Market Association (LBMA) vaults and audited monthly. This positioning makes PAXG a form of “digital gold token” that can be traded like other crypto assets while remaining supported by a tangible underlying asset. PAXG was launched in September 2019 by Paxos Trust Company, a New York–based financial institution that also issues the USDP stablecoin. Paxos was founded by Charles Cascarilla and Rich Teo, both of whom have professional backgrounds in the

²³ Tether Gold official website. <https://gold.tether.to/>

²⁴ Ryan, Jack, Tether Holds an \$8 Billion Pile of Gold in a Secret Swiss Vault. Bloomberg.com (8 July 2025). <https://www.bloomberg.com/news/articles/2025-07-08/crypto-firm-tether-holds-8-billion-pile-of-gold-in-a-secret-swiss-vault>

financial sector.²⁵ On September 5, 2019, in New York, Superintendent of Financial Services Linda A. Lacewell announced that the New York State Department of Financial Services (DFS) had granted authorization to Paxos Trust Company LLC to issue a gold-backed virtual currency, the first of its kind to receive approval from DFS. DFS remains committed to fostering innovation while maintaining strong regulatory oversight to ensure the safety and stability of both consumers and the rapidly evolving financial services sector.²⁶ Below figure showed that the market cap of tokenized gold has climbed to \$3.9 billion, led by Tether Gold (XAUT) at roughly \$2.1 billion and Pax Gold (PAXG) at \$1.3 billion. The chart shows a clear, uninterrupted rise since mid-2021, with the category growing nearly 50× in that period.

Figure 3

Market Capitalization of Tokenized Gold Jan 2021 – Sep 2025



Source: X, ambcrypto.com

Figure 3 illustrates the rapid growth of tokenized commodities over time, showing a steady increase followed by a sharp acceleration in recent periods. This upward trend reflects rising institutional and market interest in using blockchain technology to represent real-world commodities such as gold and other tangible assets in tokenized form. The gradual early growth indicates an initial adoption phase, while the later surge suggests increasing confidence, regulatory clarity, and broader use of tokenized

²⁵ Paxos official website. <https://www.paxos.com/pax-gold>

²⁶ Superintendent of Financial Services Linda A. Lacewell Announces Approval of First Gold-Backed Virtual Currency in New York State, Paxos Trust Company LLC, an Existing DFS Licensee, is Now Authorized to Offer “PAX Gold” (5 September 2019) https://www.dfs.ny.gov/reports_and_publications/press_releases/pr1909051

commodities as investable instruments. Overall, figure 3 highlights how tokenization is transforming traditionally illiquid or operationally complex commodity markets into more accessible, transparent, and digitally integrated financial assets.

On the other hand, there are several challenges of tokenized gold which may not be the issue in the near future, however as of today, one of the key challenges is the absence of a universally recognized digital gold standard. Fragmentation further compounds this issue, as different gold tokens representing the same ounce of gold are often not directly interchangeable, requiring users to rely on exchanges that list multiple tokens or to convert them through fiat currency. Another fundamental challenge lies in the verification process, verifying that a gold token genuinely corresponds to a provable quantity of physical gold held in custody. No combination of legal frameworks, reserve requirements, or audits can provide an absolute guarantee, hence established institutions such as HSBC can leverage their reputations to enhance confidence in tokenized gold products, and certain jurisdictions such as London or Switzerland carry inherent credibility, trust ultimately depends on market acceptance. The prominence of Tether as the largest issuer of gold-backed tokens today illustrates this reality: trust persists as long as the market is willing to grant it. Finally, there are technological challenges related to scalability and interoperability. Like other fungible tokens, NFTs, and stablecoins, digital gold tokens would be more economically efficient if deployed on fast, scalable public blockchains taking example of Paxos Gold (PAXG) operate as ERC-20 tokens on the Ethereum network, while HSBC's Gold Token is issued on a private distributed ledger. This diversity of platforms, while innovative, also highlights ongoing challenges in standardization, liquidity, and cross-network usability for tokenized gold.²⁷

From an economic perspective, tokenized gold improves allocative efficiency by lowering barriers to entry through fractional ownership, reducing storage and custody costs, and enabling faster, cross-border transfers compared to physical gold. Blockchain-based settlement also minimizes reconciliation costs and enhances market liquidity. In Posner's framework, these efficiency gains justify legal recognition of tokenized gold as a legitimate financial instrument, provided the legal system supports predictable enforcement of rights. Regulatory challenges arise precisely where efficiency gains introduce new risks. The most significant challenge is information asymmetry, a central

²⁷ Southurst, Jon, "Gold is back: Could digital gold tokens be the new stablecoins?" Coin Geek (1 December 2025) <https://coingeek.com/gold-is-back-could-digital-gold-tokens-be-the-new-stablecoins/>

concern in Posner's theory. Token holders must rely on issuers' representations that each token is fully backed by physical gold held in custody. Even with audits and reserve attestations, the possibility of misrepresentation cannot be eliminated. Under Posner's analysis, regulation should therefore aim to minimize verification costs and reduce opportunistic behavior, rather than attempt absolute certainty, which would be economically inefficient. Custody and legal enforceability present further challenges. In the case of PAXG, authorization by the New York State Department of Financial Services (DFS) reflects a regulatory choice to reduce uncertainty by placing the issuer under a trusted legal regime, hence lowers transaction costs by increasing market confidence and reducing the need for private due diligence. By contrast, Tether Gold operates under a more market-driven trust model, where acceptance depends largely on brand reputation and liquidity. Another regulatory issue concerns interoperability and market fragmentation. Multiple gold tokens representing the same underlying asset but issued on different platforms can reduce liquidity and raise switching costs. Posner's efficiency framework suggests that legal standards promoting disclosure, fungibility, and secondary market clarity would improve market outcomes without overregulating the technology itself.

3.3 Assessing OJK's role in the future of tokenization in the business and financial sectors in Indonesia

As digital systems continue to evolve, innovations in fund mobilization and financing mechanisms have advanced significantly, particularly through the emergence of financial technology (fintech) solutions such as peer-to-peer lending and crowdfunding. Referring to the Indonesian Fintech Lending Association, the early milestones of fintech development became evident in the early 2000s, marked by a 17.32 percent increase in the use of e-banking services compared to the previous period, with transaction values reaching IDR 6,447 trillion in 2014.²⁸ In Indonesia, the fintech industry is predominantly composed of payment services, lending, crowdfunding, aggregators, and various other business models that leverage digital platforms. These models expand access to financing for individuals and micro, small, and medium

²⁸ Asosiasi Fintech Pendanaan Bersama Indonesia (AFPI). Sejarah Perkembangan Fintech di Indonesia <https://afpi.or.id/articles/detail/sejarah-perkembangan-fintech-di-indonesia>

enterprises (MSMEs), enabling direct access to funding without reliance on traditional financial intermediaries. The growth of the fintech financing industry has had a substantial impact on the national financial landscape, with total financing disbursement reaching IDR 829.18 trillion as of July 2023. Fintech serves as an alternative financing channel for MSMEs and supports the capitalization of micro and small enterprises in a more efficient and timely manner, under the active supervision of the Financial Services Authority (Otoritas Jasa Keuangan/OJK). Within this evolving context, fintech in general was covering what is called off-chain assets, as they are not supported by blockchain technology, while what is called on-chain assets are assets with digital representations on blockchain-based systems.

Previously, the blockchain and crypto-asset ecosystem in Indonesia operated in parallel with the fintech sector, with crypto assets regulated by the Commodity Futures Trading Regulatory Agency (Bappebti) and fintech activities overseen by the Financial Services Authority (OJK). However, following the enactment of Law No. 4 of 2023 on the Development and Strengthening of the Financial Sector (UU PPSK) in January 2023, regulatory authority over crypto assets was transferred to OJK, effective January 2024. Under Chapter XVI of UU PPSK, Article 213 introduces the concept of Inovasi Teknologi Sektor Keuangan (ITSK), which in English is commonly rendered as Financial Sector Technology Innovation (FSTI). Article 213 generally defines ITSK as activities involving the development of new business processes, business models, financial instruments, and financial services through the application of technology within the financial sector. In regulatory and academic discourse in Indonesia, ITSK is typically translated as either Financial Sector Technology Innovation or Financial Technology Innovation in the Financial Sector, depending on the context. This provision serves as the legal foundation for OJK's authority to regulate emerging technologies, including crypto assets, blockchain-based services, and real-world asset tokenization, under an integrated financial regulatory framework.

Article 217 provides that the regulation, supervision, and implementation of Financial Sector Technology Innovation (ITSK) must be conducted through coordinated efforts between the Financial Services Authority (OJK) and Bank Indonesia. This coordination is designed to ensure regulatory coherence, maintain financial system stability, and enable effective oversight of technological innovation within the financial sector. The article further designates the regulatory sandbox as a central mechanism

within this coordination framework, allowing ITSK activities to be tested in a controlled environment prior to wider market deployment. In this context, the scope of regulation under POJK No. 13/POJK.02/2018 was relatively limited, as it focused broadly on digital financial innovation with an emphasis on conventional fintech business models such as digital payments, peer-to-peer lending platforms, and financial aggregators. While it introduced the regulatory sandbox into Indonesia's fintech landscape, the sandbox functioned primarily as a voluntary and experimental mechanism aimed at assessing business feasibility and managing basic consumer risks.

By contrast, POJK No. 3 of 2024 explicitly regulates Financial Sector Technology Innovation (ITSK) in accordance with the mandate of the UU PPSK. Its scope extends to blockchain-based financial services, crypto assets, tokenization including that of real-world asset (RWA) and hybrid models that integrate traditional financial institutions with digital infrastructure. Through this regulation, OJK significantly expands and clarifies the regulatory perimeter, making it more suitable for advanced and complex use cases such as tokenized assets. Moreover, under POJK No. 3 of 2024, the regulatory sandbox is transformed into a mandatory regulatory gateway for ITSK activities. The outcomes of sandbox participation now determine eligibility for licensing, compliance classification, and integration into the formal financial system. Consistent with Article 217 of the UU PPSK, the sandbox is administered through coordinated oversight between OJK and Bank Indonesia. As a result, the sandbox in Indonesia has evolved from a preliminary testing tool into a formal regulatory filter—serving not only to encourage innovation and protect consumers during early experimentation, but also to safeguard financial system stability and integrate technological innovation into the national financial architecture.

The prominence of RWAs continues to rise, the total value locked in RWA protocols, excluding fiat-backed stablecoins, has increased from approximately ~\$2 billion at the start of 2023 to nearly \$8 billion today.²⁹ The market value of asset tokenization is projected to grow substantially, reaching approximately USD 16.1 trillion by 2030, equivalent to around 10 percent of global gross domestic product (GDP).³⁰ Furthermore, the implementation of tokenization has demonstrated significant growth

²⁹ Overview of Real World Assets on Solana <https://www.reflexivityresearch.com/all-reports/overview-of-real-world-assets-on-solana>

³⁰ BCG, ADDX report: Asset tokenization to grow 50x into US\$16 trillion opportunity by 2030 (22 Sept 2022). <https://addx.co/insights/bcg-addx-report-asset-tokenization-to-grow-50x-into-us-16-trillion-opportunity-by-2030/>

across multiple jurisdictions worldwide. Countries with innovation-friendly regulatory frameworks and advanced digital financial infrastructure have emerged as pioneers in leveraging blockchain technology to transform the ownership and trading of traditional assets. Blockchain-based business applications and the implementation of an asset management system in Indonesia must adhere to Presidential Regulation No. 132 of 2022 concerning the Electronic-Based Government System (SPBE) provides important guidance regarding the SPBE architecture, encompassing basic components and architectural domains involving business processes, data and information, infrastructure, applications, security, and SPBE services.³¹

Despite the proactive measures from OJK to regulate and supervise RWA tokenization, and the potential that Indonesia has in the development of real-world asset (RWA) tokenization, particularly across the property, corporate debt, and infrastructure sectors, the issue to address is the custody of the underlying real-world assets that support digital token issuance. RWA tokens represent legally enforceable claims over physical assets, raising critical questions regarding which parties lawfully hold, manage, and guarantee the existence and validity of those underlying assets. In the absence of a clear and legally enforceable custodial framework, there is a risk that tokens may not be fully backed by real assets, creating vulnerabilities to misuse, illicit practices, and legal disputes.

In advanced jurisdictions such as the United States and Europe, this challenge is typically addressed through licensed custodians or dedicated legal entities, such as Special Purpose Vehicles (SPVs), that legally hold the underlying assets. In Singapore, regulators such as the Monetary Authority of Singapore (MAS) require a clear legal linkage between the token and the underlying asset, whether through qualified custodians or through smart contracts combined with manual verification of the physical assets. These approaches demonstrate that custody is not merely a technological issue, but a matter of legal enforceability and institutional trust. Nevertheless, OJK has set forth regulatory sandbox as one of the initial steps toward developing a dedicated legal framework that governs the legal linkage between physical assets and their digital token representations.³²

³¹ Setyawan, Faizal Gani; Kabetta, Herman; Girinoto, “Kajian Implementasi Blockchain dalam Sistem Manajemen Aset pada Gudang Logistik Instansi XYZ” *Jurnal Info Kripto*, Volume 18 Nomor 2 (2024)

³² Beyond Infinity:Tokenisasi Real World Assets (RWA). Edisi 02. OJK Fintech Center. (Juni 2025)

The transformation of OJK's regulatory sandbox from a voluntary testing mechanism into a mandatory regulatory filter for ITSK is coherent with Richard Posner's wealth-maximization theory of law. Posner conceptualizes economic efficiency as the allocation of resources in a manner that maximizes aggregate willingness to pay, thereby increasing overall social wealth. By functioning as an ex ante screening mechanism, the sandbox reduces information asymmetry, filters out inefficient or high-risk innovations, and lowers future enforcement and transaction costs. Rather than constraining innovation, such regulatory filtering enhances allocative efficiency by directing capital and market participation toward legally enforceable, transparent, and economically viable financial technologies. In this sense, adaptive and efficiency-oriented regulation operates not as a barrier to innovation, but as a legal infrastructure that enables markets to function more efficiently.

To better contextualize Indonesia's regulatory approach to real-world asset (RWA) tokenization, it is necessary to compare it with frameworks adopted in other jurisdictions. Different countries have responded to tokenization using distinct regulatory models shaped by the maturity of their financial markets, institutional structures, and risk tolerance toward technological innovation. A comparative perspective is essential for identifying best practices, particularly with respect to legal recognition of digital assets, custody arrangements for underlying assets, and the role of regulators in balancing innovation with financial stability.

Tabel 1

Comparative Regulatory Approaches to Real-World Asset (RWA) Tokenization

Jurisdiction	Primary Regulator(s)	Regulatory Approach	Key Features	Relevance to Indonesia
Indonesia	OJK, Bank Indonesia	Integrated financial regulation under UU P2SK with mandatory regulatory sandbox	ITSK framework; regulatory sandbox as gateway; supervised; evolving custody rules	Emerging model emphasizing prudential oversight, experimentation, and financial stability
United Arab Emirates (UAE)	VARA, ADGM	Specialized virtual asset regulation	Clear licensing regimes; strong investor protection; active real estate tokenization	Demonstrates benefits of sector-specific regulators and legal certainty
Germany	BaFin	Statutory recognition of blockchain-based securities	Electronic Securities Act (eWpG); full legal recognition of tokenized securities	Illustrates formal legal enforceability of on-chain assets

Jurisdiction	Primary Regulator(s)	Regulatory Approach	Key Features	Relevance to Indonesia
Switzerland	FINMA	Principles-based, innovation-friendly regulation	Clear token classification; DLT Act; strong legal certainty; tax incentives	Shows effectiveness of flexible, technology-neutral regulation
Singapore	MAS	Risk-based regulation with strong custody requirements	Clear linkage between tokens and underlying assets; licensed custodians; pilot projects	Highlights importance of custody, legal linkage, and institutional trust

The comparative table demonstrates that, despite variations in regulatory design, successful RWA tokenization frameworks share several core principles, including legal certainty, robust custody mechanisms, and risk mitigation through transparency and oversight. Indonesia's integrated approach, led by the Financial Services Authority (OJK) in coordination with Bank Indonesia, emphasizes prudential supervision through a mandatory regulatory sandbox. While this model differs from jurisdictions such as Germany, which provides explicit statutory recognition of tokenized securities, or the United Arab Emirates, which employs specialized virtual asset regulators, Indonesia's framework reflects a cautious yet adaptive regulatory strategy. With further clarification of custody arrangements and legal linkages between tokens and underlying assets, Indonesia's approach has the potential to evolve into a credible and sustainable regulatory model that supports innovation while safeguarding financial system stability and promoting broader financial inclusion.

4 CONCLUSIONS

Indonesia has demonstrated a progressive adaptation to technological developments in the business and financial sectors, particularly in relation to blockchain innovation. Cross-institutional collaboration among Bappebti, the Financial Services Authority (OJK), Bank Indonesia (BI), and other stakeholders reflects a coordinated regulatory response to the evolving digital financial ecosystem. This study contributes theoretically by positioning real-world asset (RWA) tokenization as a bridge between traditional asset theory and decentralized digital finance, highlighting how legally enforceable rights can be transformed into programmable financial instruments. The growing regulatory and market focus on bonds, real estate, and especially gold as a

traditional safe-haven asset underscores the relevance of asset-based tokenization within Indonesia's financial architecture. By engaging in regulatory learning and cooperation with foreign jurisdictions, Indonesia is well positioned to develop a credible, inclusive, and sustainable framework for RWA tokenization, while simultaneously strengthening digital financial literacy and advancing financial inclusion.

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Authors' Contribution

All authors contributed equally to the development of this article.

Data availability

All datasets relevant to this study's findings are fully available within the article.

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